in IoT startups with a cumulative investment of more than sixty million dollars since 2014. The top four areas of funding are: lifestyle/wearables, embedded computing, industrial Internet and connected homes.

Today, various operations of the city such as traffic, electricity, water, waste management, and much more are done on IoT platforms. Samsung R&D Institute in Bengaluru, Nasscom, Intel and L&T together have announced the formation of Open Connectivity Foundation (OCF) India Ecosystem Task Force to increase awareness about global IoT standards and their benefits for the Indian IoT industry.

IoT based systems for water supply and water recycling

Increasing dependence on ground water as a reliable source of water in rural areas of India has resulted in unlimited extraction of water. Without considering the research of aquifers as well as other environmental factors, ground water source may be threatened. IoT based systems can even help to utilise waste water at malls, flats, etc. Not less than 380,000 litres of water is recycled at Orion mall, Bengaluru every day based on an IoT based system.

Greenvironment India is an environmental engineering company incubated at IIT Chennai, offering smart water and waste water management solutions for residential companies/ apartments, commercial establishments, institutions and industries.

IoT based smart sensor solutions on the cloud can monitor pH, Flow, DO (dissolved oxygen), ORP (oxidation reduction potential), TDS (total dissolved salts), level, pressure, and temperature of water during water supply. It can keep automatic records of the above parameters and send alerts over email or SMS.

Smart water management systems in multi-specialty hospitals are helping to reduce hundred litres/person/head resulting in 20-30 per cent water conservation. Today, in 172 apartments of Chennai, dependency on an external supply of water is greatly reduced by producing good-quality recycled water for 55 per cent of the communities living here using an efficient IoT based system. This has resulted in saving of thirty per cent of overall water costs.

Figure1: Multi-sensor, multi-parameter, multichannel water analysing instrument to measure pH, residual chlorine, ORP, ozone, etc (Credit: *processinstruments.co.uk*)

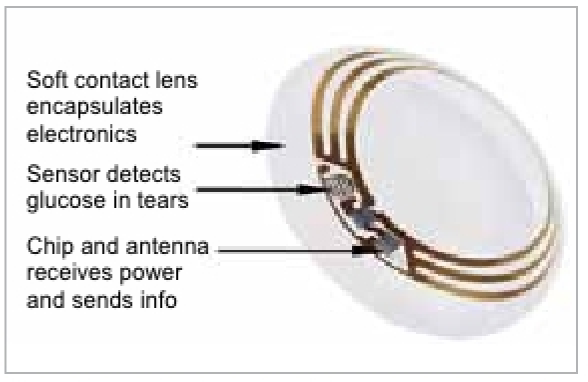

Figure 2: Google Novartis diabetes smart contact lens to improve vision (Credit: *pinterest.com*)

Effects of IoT based wearables in the healthcare system

The market size for wearables had reached US$ 25 billion by the end of 2019. Currently, wearable technologies are considered as an integral part of IoT based projects and include smartwatches, fitness bands, and even AR/VR headsets.

The smart contact lens is one of the most anticipated medical IoT technologies. Pharmaceutical giant Novartis teamed up with Google to build this revolutionary device that promises to help patients with diabetes by measuring their glucose levels as well as assisting those with eye problems. The smart contact lens looks like a regular contact lens, but it comes with a sensor that can track the blood sugar level of the user non-invasively through the tears and correct vision in a new innovative way. The project was initially designed by Google X.

The lens contains a tiny and ultra-slim microchip that is embedded in one of its thin concave sides. Through its equally tiny antenna, it will send data about glucose measurements from the user's tears to his or her paired smartphone via installed software. This technology has got the potential to lower the cost of managing the chronic disease and encourage people to get involved in managing their health digitally.

To sum up

IoT is an emerging field of Internet technology. It is a field of research and innovation concerned with increasing the abilities of devices on the Internet scale. Ordinary rules of operating devices no longer apply here. Basically, here the Internet leaves its classical realm and enters a new world of control of devices mechanism.

The scientific community, policy makers, apex bodies and funding agencies are continuously striving to progress in various areas of IoT. In the last two decades, India has achieved many milestones in the areas of IoT and many are yet to be achieved. END

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The article was originally published in the July 2020 issue of Electronics For You.